

TrES-3b

R-1770

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210517 · created 2026-08-02T22:02:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459351.8806 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.051
Transit depth (Rp/Rs)^2	3.59 +/- 1.93 [%]
Orbital Inclination (inc)	80.29 +/- 2.09
Airmass coefficient 1 (a1)	0.9664 +/- 0.0083
Airmass coefficient 2 (a2)	0.034 +/- 0.035
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#2 - [301, 60]
Optimal Aperture	3.73
Optimal Annulus	11.82
Transit Duration (day)	0.035 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.189 ± 0.051 vs 0.16309 (deviation 0.51σ)
Duration fit vs published	0.035 d vs 0.0567 d (deviation 0.83σ)
Mid-time O–C	3.1 min
Depth SNR	3.7
Residual scatter	0.86 %

Light curve

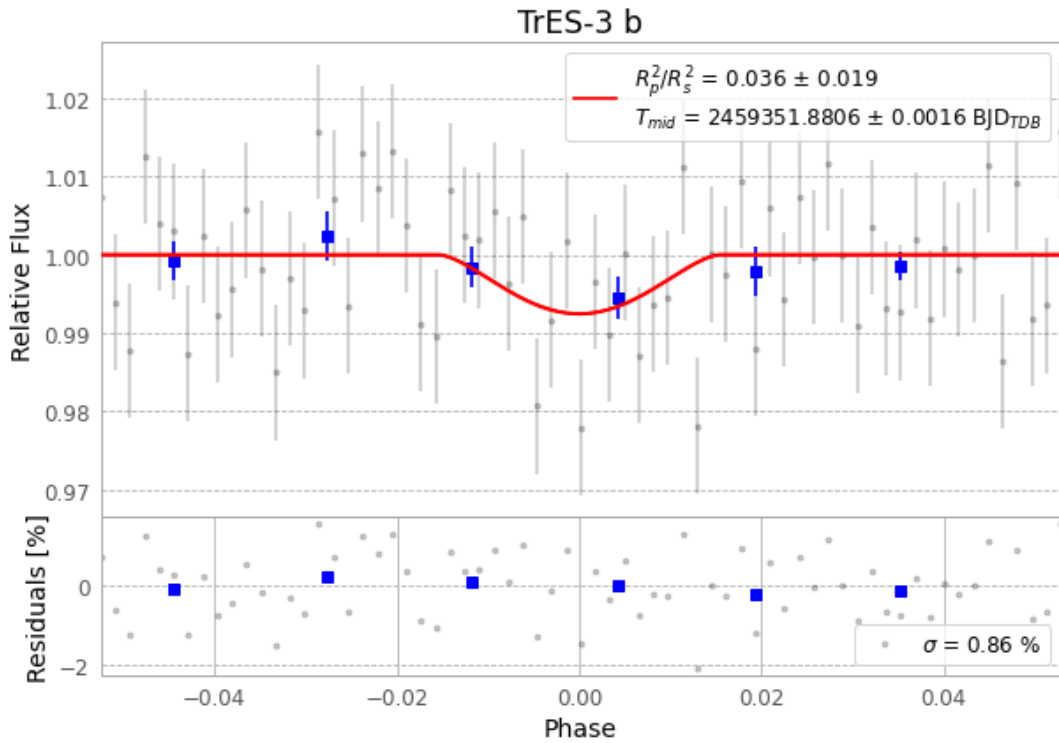


Figure 1 — FinalLightCurve_TrES-3 b_2021-05-17.png (SHA-256 ea851c6f5916883f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 afb3013ed6b1bd11...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3210517072412.FITS → TRES-3210517104213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210517.log (a2e65b3f8720...), FinalLightCurve_TrES-3 b_2021-05-17.png (ea851c6f5916...), FinalLightCurve_TrES-3 b_2021-05-17.csv (233a2cbd8bbc...)

Compute resources

Wall time 230.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:02Z.